

SQL ASSIGNMENT
PROJECT : HOSPITAL MANAGEMENT

SUBMITTED BY
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DATA BASE

Use database hospital;

Assignment Queries:

1. List all patients with their name, age, gender, and phone number.

use hospital;

create database hospital;

create table patients(

patient_id int auto_increment primary key,

Name varchar(400),

Age int,

Gender enum('Female','Male','Others'),

phone_number bigint not null,

createdat timestamp default current_timestamp

);

insert into patients(Name,Age,Gender,Phone_number,city) values

('divya',30,'Female',2345678901,'bangalore'),

('praveen',23,'Male',2345678234,'mumbai'),

('sunil',18,'Male',8905678901,'jaipur');

select * from patients;

update patients set name = 'amit' where patient_id = 2;

alter table patients add column city varchar(250);

	patient_id	Name	Age	Gender	phone_number	createdat
1		divya	30	Female	2345678901	2026-01-05 18:00:13
2		praveen	23	Male	2345678234	2026-01-05 18:00:13
3		sunil	18	Male	8905678901	2026-01-05 18:00:13
	NULL	NULL	NULL	NULL	NULL	NULL

2. List all doctors with their name and specialization

create table doctors(

Doctor_id int auto_increment primary key ,

Name varchar(250) not null,

Age int,

Gender enum('Female','Male','Others'),

```

phone_number bigint not null,

specialization varchar(250),

createdat timestamp default current_timestamp

);

insert into doctors(Name,Age,Gender,phone_number,specialization) values
('kala',55,'Female',9916300184,'Cardiologists'),

('sundar',35,'Male',9916302345,'Neurologists'),

('surya',25,'Male',9916144575,'ENT Specialists');

select * from doctors;

SELECT name, specialization FROM doctors;

update doctors set specialization = 'Orthopedics' where doctor_id = 1;

update doctors set specialization = 'dermatology' where doctor_id = 2;

update doctors set name = 'Dr. Sharma' where doctor_id = 1;

```

	Doctor_id	Name	Age	Gender	phone_number	specialization	createdat
▶	1	kala	55	Female	9916300184	Cardiologists	2026-01-05 18:54:42
◀	2	sundar	35	Male	9916302345	Neurologists	2026-01-05 18:54:42
	3	surya	25	Male	9916144575	ENT Specialists	2026-01-05 18:54:42
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL

	name	specialization
▶	kala	Cardiologists
	sundar	Neurologists
	surya	ENT Specialists

3. Show all appointments scheduled for today.

```

create table appointments(

Appointment_id int auto_increment primary key,

status enum ('scheduled','cancelled','completed'),

date_time datetime,

fees int,

patient_id int,

Doctor_id int,

createdat timestamp default current_timestamp,

foreign key(patient_id) references patients(patient_id),

foreign key(Doctor_id) references doctors(Doctor_id)

);

```

insert into appointments(patient_id,doctor_id,status,date_time,fees) values

(1,1,'completed','2023-03-26 23:22:12',1000),

(2,2,'cancelled','2023-06-08 12:22:12',400),

(3,3,'scheduled','2026-04-16 19:22:12',1000);

select * from appointments;

select * from appointments where date(date_time)=curdate();

	Appointment_id	status	date_time	fees	patient_id	Doctor_id	createdat
▶	1	completed	2023-03-26 23:22:12	1000	1	1	2026-01-05 19:00:21
	2	cancelled	2023-06-08 12:22:12	400	2	2	2026-01-05 19:00:21
	3	scheduled	2026-04-16 19:22:12	1000	3	3	2026-01-05 19:00:21
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL

	Appointment_id	status	date_time	fees	patient_id	Doctor_id	createdat
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL

4. Find the total number of patients in the system.

select count(*) as total_patients from patients;

	total_patients
▶	3

5. Find the total number of doctors in the hospital.

select count(*) as total_doctors from doctors;

	total_doctors
▶	3

6. List all bills with status 'Paid'.

create table bills(

bill_id int auto_increment primary key,

patient_id int,

bil_amount bigint,

bill_status enum('paid','unpaid'),

date_time datetime,

createdat timestamp default current_timestamp,

foreign key(patient_id) references patients(patient_id)

);

insert into bills(patient_id,bil_amount,bill_status,date_time) values

(1,23000,'paid','2023-04-09 23:33:16'),

(2,28000,'unpaid','2023-10-19 12:33:16'),

```
(3,78000,'paid','2023-09-09 12:33:16');
```

```
select * from bills;
```

```
SELECT * FROM bills WHERE bill_status = 'Paid';
```

	bill_id	patient_id	bil_amount	bill_status	date_time	createdat
▶	1	1	23000	paid	2023-04-09 23:33:16	2026-01-05 19:35:43
	2	2	28000	unpaid	2023-10-19 12:33:16	2026-01-05 19:35:43
	3	3	78000	paid	2023-09-09 12:33:16	2026-01-05 19:35:43
•	NULL	NULL	NULL	NULL	NULL	NULL

	bill_id	patient_id	bil_amount	bill_status	date_time	createdat
▶	1	1	23000	paid	2023-04-09 23:33:16	2026-01-05 19:35:43
	3	3	78000	paid	2023-09-09 12:33:16	2026-01-05 19:35:43
•	NULL	NULL	NULL	NULL	NULL	NULL

7. List all bills with status 'Unpaid'

```
select * from bills where bill_status = 'unpaid';
```

	bill_id	patient_id	bil_amount	bill_status	date_time	createdat
▶	2	2	28000	unpaid	2023-10-19 12:33:16	2026-01-05 19:35:43
•	NULL	NULL	NULL	NULL	NULL	NULL

8. Display all medications prescribed to patient named 'Amit'.

```
use hospital;
```

```
create table medicines(
```

```
medicines_id int auto_increment primary key,
```

```
name varchar(250),
```

```
price int,
```

```
status enum('delivered','not delivered','issue'),
```

```
time_date datetime,
```

```
patient_id int,
```

```
doctor_id int,
```

```
createdat timestamp default current_timestamp,
```

```
foreign key(patient_id) references patients(patient_id),
```

```
foreign key(doctor_id) references doctors(doctor_id)
```

```
);
```

```
insert into medicines (name,price,status,time_date,patient_id,doctor_id) values
```

```
('paracetamol',150,'delivered','2023-04-09 23:33:16',1,1),
```

```
('dart',130,'not delivered','2026-01-11 22:09:13',2,2),
```

```
('cipla',150,'delivered','2026-09-13 11:09:23',3,3);
```

select * from medicines;

select p.name as patient_name, m.name as medicine_name from patients p join medicines on p.patient_id = m.patients_id where p.name = 'amit';

select m.* from medicines m join patients p on m.patient_id = p.patient_id where p.name = 'amit';

	medication_id	name	price	status	time_date	patient_id	doctor_id	createdat
▶	1	paracetamol	150	delivered	2023-04-09 23:33:16	1	1	2026-01-05 20:08:03
⊗	2	dart	130	not delivered	2026-01-11 22:09:13	2	2	2026-01-05 20:08:03
	3	cipla	150	delivered	2026-09-13 11:09:23	3	3	2026-01-05 20:08:03
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

	medicines_id	name	price	status	time_date	patient_id	doctor_id	createdat
▶	2	dart	130	not delivered	2026-01-11 22:09:13	2	2	2026-01-06 09:36:21

9. Retrieve the details of doctors who specialize in 'Orthopedics'.

select * from doctors where specialization = 'Orthopedics';

	Doctor_id	Name	Age	Gender	phone_number	specialization	createdat
▶	1	kala	55	Female	9916300184	Orthopedics	2026-01-05 20:05:47
⊗	NULL	NULL	NULL	NULL	NULL	NULL	NULL

10. List the names of patients older than 60.

select * from patients where Age < 60;

	patient_id	Name	Age	Gender	phone_number	createdat	city
▶	1	divya	30	Female	2345678901	2026-01-05 20:05:22	NULL
	2	amit	23	Male	2345678234	2026-01-05 20:05:22	NULL
	3	sunil	18	Male	8905678901	2026-01-05 20:05:22	NULL
	4	divya	30	Female	2345678901	2026-01-06 19:16:01	bagalore
	5	praveen	23	Male	2345678234	2026-01-06 19:16:01	mumbau

11. Show all appointments for doctor named 'Dr. Sharma'.

select a.* from doctors d join appointments a on d.Doctor_id=a.Doctor_id where d.name = 'Dr. Sharma';

	Appointment_id	status	date_time	fees	patient_id	Doctor_id	createdat
▶	1	completed	2023-03-26 23:22:12	1000	1	1	2026-01-05 20:05:47

12. Show the phone numbers of patients who had appointments in the last 7 days.

select distinct phone_number from patients p join appointments a on p.patient_id=a.patient_id where date_time >= curdate() - interval 7 day;

	phone_number
▶	8905678901

13. Find all appointments that were cancelled.

select * from appointments where status = 'cancelled';

	Appointment_id	status	date_time	fees	patient_id	Doctor_id	createdat
▶	2	cancelled	2023-06-08 12:22:12	400	2	2	2026-01-05 20:05:47
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

14. Show all unique specializations available in the hospital.

select distinct specialization from doctors;

	specialization
▶	Orthopedics
	dermatology
	ENT Specialists

15. List patients who live in a specific city (e.g., 'Mumbai').

select * from patients where city ='bangalore';

	patient_id	Name	Age	Gender	phone_number	createdat	city
▶	7	divya	30	Female	2345678901	2026-01-06 19:19:53	bangalore
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

16. Count the number of appointments each doctor has had.

SELECT Doctor_id, COUNT(*) AS total_appointments FROM appointments GROUP BY Doctor_id;

	Doctor_id	total_appointments
▶	1	1
	2	1
	3	1

17. Find the number of appointments made by each patient.

select patient_id, count(*) as num_of_appointments from appointments group by patient_id;

	patient_id	num_of_appointments
▶	1	1
	2	1
	3	1

18. Calculate the total amount billed to each patient

select patient_id, sum(bil_amount) as total_billed_amont from bills group by patient_id;

	patient_id	total_billed_amont
▶	1	23000
	2	28000
	3	78000

19. List all doctors who have more than 5 appointments.

select Doctor_id from appointments group by Doctor_id having count(*) > 5;

Doctor_id

20. Show the patients who have been prescribed more than 2 medications.

select patient_id from medicines group by patient_id having count(*) > 2;

patient_id

21. Retrieve the latest appointment for each patient

select * from appointments a where date_time =(select max(date_time)from appointments a2 where a.patient_id=a2.patient_id);

Appointment_id	status	date_time	fees	patient_id	Doctor_id	createdat
1	completed	2023-03-26 23:22:12	1000	1	1	2026-01-05 20:05:47
2	cancelled	2023-06-08 12:22:12	400	2	2	2026-01-05 20:05:47
3	scheduled	2026-04-16 19:22:12	1000	3	3	2026-01-05 20:05:47
NULL	NULL	NULL	NULL	NULL	NULL	NULL

22. Display patients who have unpaid bills greater than Rs.10,000.

select patient_id from bills where bill_status ='unpaid' group by patient_id having sum(bil_amount)> 10000;

patient_id
2

23. List patients who have had appointments with more than one doctor

select patient_id from appointments group by patient_id having count(distinct Doctor_id)>1;

patient_id

24. Show all patients who don't have any upcoming appointments.

select * from patients where patient_id not in(select patient_id from appointments where date_time > now());

patient_id	Name	Age	Gender	phone_number	createdat	city
1	divya	30	Female	2345678901	2026-01-05 20:05:22	NULL
2	amit	23	Male	2345678234	2026-01-05 20:05:22	NULL
4	divya	30	Female	2345678901	2026-01-06 19:16:01	bagalore
5	praveen	23	Male	2345678234	2026-01-06 19:16:01	mumbau
6	sunil	18	Male	8905678901	2026-01-06 19:16:01	jaipur
7	divya	30	Female	2345678901	2026-01-06 19:19:53	bangalore
8	praveen	23	Male	2345678234	2026-01-06 19:19:53	mumbai
9	sunil	18	Male	8905678901	2026-01-06 19:19:53	jaipur
NULL	NULL	NULL	NULL	NULL	NULL	NULL

25. Find the total revenue generated in the current month.

select sum(bil_amount) as total_revenue from bills where month(date_time)=month(curdate());

total_revenue
NULL

26. Show the number of medications prescribed to each patient:

select patient_id , count(*) as total_medicines from medicines group by patient_id;

	patient_id	total_medicines
▶	1	1
	2	1
	3	1

27. List all patients who were treated by doctors specialized in 'Dermatology'.

select distinct p.* from patients p join appointments a on p.patient_id=a.patient_id join doctors d on a.Doctor_id=d.Doctor_id where d.specialization = 'dermatology';

	patient_id	Name	Age	Gender	phone_number	createdat	city
▶	2	amit	23	Male	2345678234	2026-01-05 20:05:22	NULL

28. Show appointments grouped by their status (Scheduled, Completed, Cancelled).

select status, count(*) as total from appointments group by status;

	status	total
▶	completed	1
	cancelled	1
	scheduled	1

29. Display the average bill amount for each patient.

select patient_id, avg(bil_amount) as avg_bill_amount from bills group by patient_id;

	patient_id	avg_bill_amount
▶	1	23000.0000
	2	28000.0000
	3	78000.0000

30. Find doctors who have not had any appointments in the past 30 days.

select * from doctors where Doctor_id not in (select Doctor_id from appointments where date_time >= curdate() - interval 30 day);

	Doctor_id	Name	Age	Gender	phone_number	specialization	createdat
▶	1	Dr. Sharma	55	Female	9916300184	Orthopedics	2026-01-05 20:05:47
	2	sundar	35	Male	9916302345	dermatology	2026-01-05 20:05:47
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL

31. Find the top 3 doctors with the most completed appointments

SELECT doctor_id, COUNT(*) FROM appointments WHERE status = 'Completed' GROUP BY doctor_id ORDER BY COUNT(*) DESC LIMIT 3;

	doctor_id	COUNT(*)
▶	1	1

32. List patients who have visited doctors of more than one specialization

select a.patient_id from appointments a join doctors d on a.Doctor_id=d.Doctor_id group by patient_id having count(distinct d.specialization)> 1;

patient_id

33. Generate a monthly report of total billed and paid amounts

select month(date_time) as month, year(date_time) as year,

sum(bil_amount) as total_billed, sum(case when bill_status = 'paid' then bil_amount else 0 end) as

total_paid from bills group by year (date_time) ,month(date_time);

	month	year	total_billed	total_paid
▶	4	2023	23000	23000
	10	2023	28000	0
	9	2023	78000	78000

34. Identify patients who have never had an appointment.

select * from patients where patient_id not in(select distinct patient_id from appointments);

	patient_id	Name	Age	Gender	phone_number	createdat	city
▶	4	divya	30	Female	2345678901	2026-01-06 19:16:01	bagalore
	5	praveen	23	Male	2345678234	2026-01-06 19:16:01	mumbau
	6	sunil	18	Male	8905678901	2026-01-06 19:16:01	jaipur
	7	divya	30	Female	2345678901	2026-01-06 19:19:53	bangalore
	8	praveen	23	Male	2345678234	2026-01-06 19:19:53	mumbai
	9	sunil	18	Male	8905678901	2026-01-06 19:19:53	jaipur
*	1001	1001	1001	1001	1001	1001	1001

35. Identify patients who have never been prescribed any medication

select * from patients where patient_id not in(select distinct patient_id from medicines);

patient_id

36. Show patients with total unpaid bills exceeding Rs.50,000

select patient_id from bills where bill_status ='unpaid' group by patient_id having sum(bil_amount)>50000;

	Doctor_id	patient_count
▶	1	1

37. Display the doctor who has treated the most unique patients

select Doctor_id,count(distinct patient_id)as patient_count from appointments group by Doctor_id order by patient_count desc limit 1;

patient_id

38. List patients who had appointments on consecutive days

select distinct a1.patient_id from appointments a1 join appointments a2 on a1.patient_id = a2.patient_id where datediff(a1.date_time, a2.date_time) =1;

patient_id

39. For each doctor, list their name and the number of unique patients they've treated

select d.name, count(distinct a.patient_id) as unique_patients from doctors d left join appointments a on d.Doctor_id = a.Doctor_id group by d.Doctor_id;

	name	unique_patients
▶	Dr. Sharma	1
	sundar	1
	surya	1

40. Display a report of patients, their total number of appointments, total bill amount, and number of medications prescribed.

select p.patient_id, p.name, count(distinct a.Appointment_id)as total_appointments, sum(b.bil_amount) as total_billed, count(distinct m.medicines_id) as total_medicines

from patients p

left join appointments a on p.patient_id = a.patient_id left join bills b on

p.patient_id=b.patient_id left join medicines m on p.patient_id = m.patient_id group by p.patient_id;

	patient_id	name	total_appointments	total_billed	total_medicines
▶	1	divya	1	23000	1
	2	amit	1	28000	1
	3	sunil	1	78000	1
	4	divya	0	NULL	0
	5	praveen	0	NULL	0
	6	sunil	0	NULL	0
	7	divya	0	NULL	0
	8	praveen	0	NULL	0
	9	sunil	0	NULL	0